Lecture 31 — Rank, column space, null space, and basis connections

Wed Nov 4

Last updated: October 8, 2026.

Boyd video (optional): Lecture 15 — linear independence (rank ideas) · Lecture 32 — pseudoinverse · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Strang / MIT (optional): Reading: Strang §3.5 (PDF) — four subspaces / rank
Video: 18.06 Lec 6 — column space and nullspace · 18.06 Lec 10 — four fundamental subspaces · MIT 18.06 playlist (Strang)

Reading: Boyd plus selected portions of Strang §3.5, especially rank, column space, null space, basis, and dimension. Related objectives: Lectures. Schedule.

Note: Not a full four-subspaces unit. Row space and left nullspace appear briefly as enrichment only — **not** quiz material.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, students should be able to:

1. Interpret column space and null space through $Ax = b$ and $Ax = 0$.
2. Determine rank from pivots or independent columns.
3. Find simple bases for column or null spaces.
4. Connect rank to consistency, independence, and dimension.
5. Use subspace, basis, dimension, rank, column space, and null space correctly.

Warm-up

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix}.$$

(a) Are the three columns independent? (Look carefully at columns 1 and 2.)

(b) For which b can $Ax = b$ have a solution — any b in $\mathbb{R}^3$, or only some?

Fill in

The **column space** $C(A)$ is the span of the columns of A. Solving $Ax = b$ asks: is b in _____?

The **null space** $N(A)$ is the set of solutions of $Ax =$ _____. It always contains 0; nontrivial null vectors mean dependent columns.

A **subspace** is a nonempty set closed under addition and scaling. Both $C(A)$ and $N(A)$ are subspaces.

A **basis** is a linearly independent spanning set. The **dimension** of a subspace is the number of vectors in any basis.

The **rank** of A is

$$\text{rank}(A) = _____$$

(equivalently: number of pivot columns after elimination / RREF). Then

$$\dim C(A) = \text{rank}(A), \qquad \dim N(A) = n - \text{rank}(A)$$

for an $m \times n$ matrix (number of free variables = nullity).

Reading RREF for bases.

- Pivot columns of the *original* A form a basis for $C(A)$.
- Free-variable special solutions form a basis for $N(A)$.

> **i** Enrichment (not on quizzes)
>
> Strang's other two subspaces: the **row space** $C(A^\mathsf{T})$ and **left nullspace** $N(A^\mathsf{T})$. Same rank story; we do not drill them for Quiz 9 / Midterm 2.

Worked in class

W1. Column space via $Ax = b$. Let

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Describe $C(A)$ in words. Is $b = (1, 2, 3)$ in $C(A)$? Is $b = (1, 3, 0)$?

W2. Null space. Same A. Solve $Ax = 0$. Give a basis for $N(A)$ (it may be $\{0\}$ only — say so).

W3. Rank from pivots. The RREF of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix}$$

is

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

What is $\text{rank}(A)$? Which columns of the original A form a basis for $C(A)$? Find a basis for $N(A)$.

W4. Dictionary. Match each phrase to the right idea: consistency of $Ax = b$; uniqueness of the solution when it exists; "how big is the column space?"

Your turn

Y1. Let

$$A = \begin{bmatrix} 1 & 2 \\ -1 & -2 \\ 0 & 0 \end{bmatrix}.$$

Find $\text{rank}(A)$ and a basis for $C(A)$. Is $b = (2, -2, 0)$ in $C(A)$?

Y2. For the same A, find a basis for $N(A)$. How many free variables?

Y3. True or false: if $\text{rank}(A) = n$ for an $m \times n$ matrix with $m \geq n$, then $Ax = 0$ has only the zero solution.

Y4. True or false: row space and left nullspace will appear on Quiz 9.

Answers (Your turn)

Y1. Columns: $a_2 = 2a_1$, so $\text{rank}(A) = 1$. Basis for $C(A)$: $\{(1, -1, 0)\}$ (or any nonzero multiple). Yes — $b = 2a_1$.

Y2. $Ax = 0$ means $x_1 + 2x_2 = 0$, so $x = (-2t,\, t) = t(-2, 1)$. Basis for $N(A)$: $\{(-2, 1)\}$. One free variable; $\dim N(A) = 1 = 2 - \text{rank}(A)$.

Y3. **True.** Full column rank means no free variables, so $N(A) = \{0\}$.

Y4. **False.** Enrichment only — not quiz material.